

This exam has 3 pages and 8 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

Please motivate all answers!

1. Semantic entailment (5 + 5 points)

- (a) Suppose the semantic entailment $\phi_1 \models \phi_2$ holds. Can we conclude that then the semantic entailment $(\phi_1 \rightarrow \psi) \models (\phi_2 \rightarrow \psi)$ holds? If so, show this by means of a logical argumentation or a truth table; if not, give a counterexample.
- (b) Suppose the semantic entailment $\psi_1 \models \psi_2$ holds. Can we conclude that then the semantic entailment $(\phi \rightarrow \psi_1) \models (\phi \rightarrow \psi_2)$ holds? If so, show this by means of a logical argumentation or a truth table; if not, give a counterexample.

2. Island puzzle (8 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). Suppose an islander A would say: “If I am a liar, then I am a truth speaker.”

Determine via a truth table whether A would be a truth speaker, a liar, both options are possible, or this statement is impossible on the island.

3. Conjunctive normal form (8 points)

Give the truth table of the DNF $(p \wedge q) \vee r$, and use it to construct a formula in CNF that is semantically equivalent.

4. CNF algorithm (10 points)

Apply the algorithm CNF to turn the formula $\neg(p \rightarrow q) \vee (\neg p \wedge q)$ into CNF.

5. DPLL procedure (9 points)

Apply the DPLL procedure to the CNF $(\neg p \vee \neg q) \wedge (\neg p \vee q) \wedge (p \vee \neg q)$, to check whether it is satisfiable.

6. Sets (6 + 9 points)

In this task, we consider the following set-theoretic equality:

$$(A' \cup (A \cap B'))' = A \cap B$$

- (a) Draw Venn diagrams for the two sets $(A' \cup (A \cap B'))'$ and $A \cap B$, depicting clearly which area corresponds to the set, and how the set is constructed from A and B using the fundamental set operations. Use your Venn-diagrams to conclude that the set-theoretic equality displayed above holds.
- (b) Prove the set-theoretic equality displayed above (for all sets A and B) using the rules of the algebra of sets.

7. Relations (4 + 6 + 6 points)

Consider the relations R and S , both in the set $V := \{1, 2, 3, 4\}$, given by the following matrix representations:

$$R: \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix} \quad S: \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

(rows and columns are numbered in the order 1, 2, 3, 4)

- (a) Is the relation S reflexive? transitive? symmetric? anti-symmetric? Briefly explain your answers.
- (b) Explicitly list all the elements of $R \circ S$ in the curly-bracket notation and draw the directed graph representation of $R \circ S$.
- (c) Find a relation in the set V that is *neither* symmetric *nor* anti-symmetric, and draw its directed graph representation.

8. Ordering relations (8 + 3 + 3 points)

In this task, we consider the Cartesian ordering relation in the set

$$A := \{a, b\}^3 = \{aaa, aab, aba, abb, baa, bab, bba, bbb\},$$

induced by the alphabetic order on $\{a, b\}$. Note that we write **aba** instead of $\langle a, b, a \rangle$, and so on, to simplify the notation.

- (a) Use the algorithm that you have learned to construct the Hasse diagram that represents the Cartesian ordering relation on A . Explicitly write down the sets G_x and H_x obtained in the construction, then draw the Hasse diagram.

Let B be the subset of A that is given by

$$B := \{\mathbf{aaa}, \mathbf{aab}, \mathbf{aba}, \mathbf{abb}, \mathbf{bab}\}.$$

- (b) Does the set B have a smallest element according to the Cartesian ordering relation from part (a)? If so, please give this smallest element. If not, please list all the minimal elements of the set B .
- (c) Does the set B have a largest element according to the Cartesian ordering relation from part (a)? If so, please give this largest element. If not, please list all the maximal elements of the set B .